

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

View Only

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QUESTIONS

Q1

ATM Withdrawal – Exception
Handling Debugging
10 marks

Q2

Hospital Registration –
NullPointerException
Debugging
10 marks

Q3

E-Commerce Product Search –
Array Exception Debugging
10 marks

Q4

Bank Account – Race Condition
Debugging
10 marks

Q5

Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q1

ATM Withdrawal – Exception Handling
Debugging10
marks

An ATM application is used to process cash withdrawals. The following Java program contains programming errors.

Task:

1. Run the given program and identify the errors.
2. Debug and modify the program.
3. Submit the COMPLETE corrected Java program.
4. The corrected program must handle insufficient balance, zero withdrawal, invalid input, and unexpected errors.
5. Use appropriate try-catch-finally exception handling.
6. The corrected program must execute successfully for all valid test cases.

Faulty Program:

```
import java.util.Scanner;

class ATM {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter account balance: ");
        int balance = sc.nextInt();
```

```
System.out.print("Enter withdrawal amount: ");
int amount = sc.nextInt();

try {
    int remaining = balance / amount;

    if (amount > balance) {
        System.out.println("Insufficient balance");
    } else {
        System.out.println("Withdrawal successful");
        System.out.println("Remaining balance = "
            + (balance - amount));
    }
}
catch (ArithmeticException e) {
    System.out.println("Transaction failed");
}

System.out.println("Transaction completed");
}
```

Expected correction:

The program must correctly validate the withdrawal amount before performing the transaction, handle division-by-zero and invalid input, and always display "Transaction completed" after exception handling.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.*;
2 class ATM{
3     public static void main(String[] args)
4     {
5         Scanner sc = new Scanner(System.in);
6         try{
7             int balance = sc.nextInt();
8
9             int amount = sc.nextInt();
10
11             int remaining = balance/amount
```

```
12         if (amount<=0)
13             throw new IllegalArgumentException("Invalid amount")
14         if (amount>balance)
15             System.out.println("Insufficient balance")
16         else{
17             System.out.println("Remaining balance = "+balance-amount)
18         }
19     }
20     catch (ArithmeticException e) {
21         System.out.println("Withdrawal amount cannot be zero")
22     }
23
24     catch (InputMismatchException e) {
25         System.out.println("Invalid input")
26     } catch (IllegalArgumentException e) {
27         System.out.println("Invalid transaction")
28     }
29
30     catch (Exception e) {
31         System.out.println("Transaction failed")
32     }
33
34     finally{
35         System.out.println("Transaction completed")
36     }
37
38
39     }
40 }
```



Test Input

10000
2000

Output

Remaining balance = 8000
Transaction completed

 Pending manual review

 Submitted

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